Finals Task 1. Using SQL to Clean and Analyze Data.

1. Download the dataset "Callcenter.csv"
2. Open xampp and MySql workbench
3. Create a Connection – assign any name for your connection.
4. Create a new schema named "Project"
5. S1elect the database/schema

```
-- We want to work on our databse.

USE PROJECT;
```

Selecting the database we want to work with.

6. Open the csv file in Excel and analyze the data, it should look like the one below

Here is a snapshot from the CSV file we have:

	A	B	C	D	E	F	G	H	I	J	K	L
1	id	customer_name	sentiment	csat_score	call_timestamp	reason	city	state	channel	response_time	call duration in minutes	call_center
2	DKK-57076809-w-055481-fU	Analise Gairdner	Neutral	7	10/29/2020	Billing Question	Detroit	Michigan	Call-Center	Within SLA	17	Los Angeles/CA
3	QGK-72219678-w-102139-KY	Crichton Kidsley	Very Positive		10/05/2020	Service Outage	Spartanburg	South Carolina	Chatbot	Within SLA	23	Baltimore/MD
4	GYJ-30025932-A-023015-LD	Averill Brundrett	Negative		10/04/2020	Billing Question	Gainesville	Florida	Call-Center	Above SLA	45	Los Angeles/CA
5	ZJI-96807559-i-620008-m7	Noreen Lafflina	Very Negative	1	10/17/2020	Billing Question	Portland	Oregon	Chatbot	Within SLA	12	Los Angeles/CA
6	DDU-69451719-O-176482-Fm	Toma Van der Beken	Very Positive		10/17/2020	Payments	Fort Wayne	Indiana	Call-Center	Within SLA	23	Los Angeles/CA
7	JVI-79728660-U-224285-4a	Kaylyn Emlen	Neutral	5	10/28/2020	Billing Question	Salt Lake City	Utah	Call-Center	Within SLA	25	Baltimore/MD
8	AZI-95054097-e-185542-PT	Phillipe Bowring	Neutral	8	10/16/2020	Billing Question	Tyler	Texas	Chatbot	Within SLA	31	Baltimore/MD
9	TWX-27007918-I-608789-Xw	Krysta de Tocqueville	Positive		10/21/2020	Billing Question	New York City	New York	Chatbot	Below SLA	37	Los Angeles/CA
10	XNG-44599118-P-344473-ZU	Oran Lifsey	Very Negative		10/03/2020	Billing Question	Dallas	Texas	Email	Below SLA	37	Baltimore/MD
11	RLC-64108207-Z-285141-VS	Port Inggall	Neutral		10/07/2020	Billing Question	Cincinnati	Ohio	Chatbot	Within SLA	12	Baltimore/MD
12	RJF-00263922-O-647027-TB	Ella Cristoforo	Negative		10/09/2020	Billing Question	Everett	Washington	Chatbot	Within SLA	35	Los Angeles/CA
13	ZQN-32874873-e-786499-kJ	Aubrey Surcombe	Negative		10/11/2020	Billing Question	Huntington	West Virginia	Web	Within SLA	18	Los Angeles/CA
14	JDP-35147568-w-630120-3l	Nicolle Fareweather	Very Positive		10/02/2020	Billing Question	Portland	Oregon	Call-Center	Within SLA	30	Baltimore/MD
15	DPT-56483482-P-371409-CQ	Melesa Ricardot	Positive	7	10/10/2020	Billing Question	Springfield	Massachusetts	Chatbot	Within SLA	20	Denver/CO
16	ZOV-95861398-a-333622-9r	Odell Cathesyed	Very Negative		10/06/2020	Payments	Hyattsville	Maryland	Call-Center	Below SLA	22	Baltimore/MD
17	BEJ-69711449-V-758715-cp	Dani Stanfield	Negative	4	10/18/2020	Billing Question	New York City	New York	Chatbot	Within SLA	28	Denver/CO
18	DEC-83767217-S-314070-eR	Margarette Jehaes	Negative		10/11/2020	Billing Question	Huntsville	Alabama	Email	Above SLA	36	Baltimore/MD
19	XNY-04106353-Y-318117-I9	Noni Greatrakes	Neutral		10/30/2020	Billing Question	Wichita	Kansas	Call-Center	Above SLA	37	Baltimore/MD
20	GKH-06532516-2-756137-9w	Gerik Archell	Negative		10/26/2020	Billing Question	Lansing	Michigan	Web	Within SLA	41	Baltimore/MD
21	DJU-19977844-M-356042-cQ	Tammie Bettinson	Very Negative		10/11/2020	Payments	Lansing	Michigan	Call-Center	Within SLA	9	Chicago/IL
22	ADD-82219259-r-882390-EG	Errol Follos	Neutral		10/12/2020	Billing Question	Fort Wayne	Indiana	Chatbot	Below SLA	35	Baltimore/MD
23	YOB-40492230-M-009287-T8	Nanni Doy	Negative	5	10/08/2020	Billing Question	Hayward	California	Email	Within SLA	27	Baltimore/MD
24	GZD-50459522-O-178569-D2	Sophie Kleinerman	Very Negative	2	10/03/2020	Billing Question	Santa Barbara	California	Chatbot	Within SLA	20	Chicago/IL
25	FQX-24118867-H-358169-3N	Timotheus Menlove	Negative		10/28/2020	Billing Question	Memphis	Tennessee	Call-Center	Within SLA	43	Baltimore/MD
26	SVH-32880745-c-681066-4a	Allayne Lednor	Negative		10/06/2020	Service Outage	Murfreesboro	Tennessee	Chatbot	Below SLA	41	Baltimore/MD
27	ISK-94965442-x-233388-Vz	Bethina Fazzioli	Positive		10/22/2020	Billing Question	Lubbock	Texas	Chatbot	Below SLA	45	Denver/CO
28	WUJ-90727821-m-177169-j0	Stanwood Esley	Very Negative		10/01/2020	Billing Question	New York City	New York	Call-Center	Within SLA	19	Baltimore/MD
29	PKG-51691289-6-484895-mg	Anissa Kinrade	Very Positive		10/05/2020	Payments	Oklahoma City	Oklahoma	Call-Center	Within SLA	40	Chicago/IL

our dataset in excel.

7. To load that data into a table in the database we need to create a table that will match it. Like this:

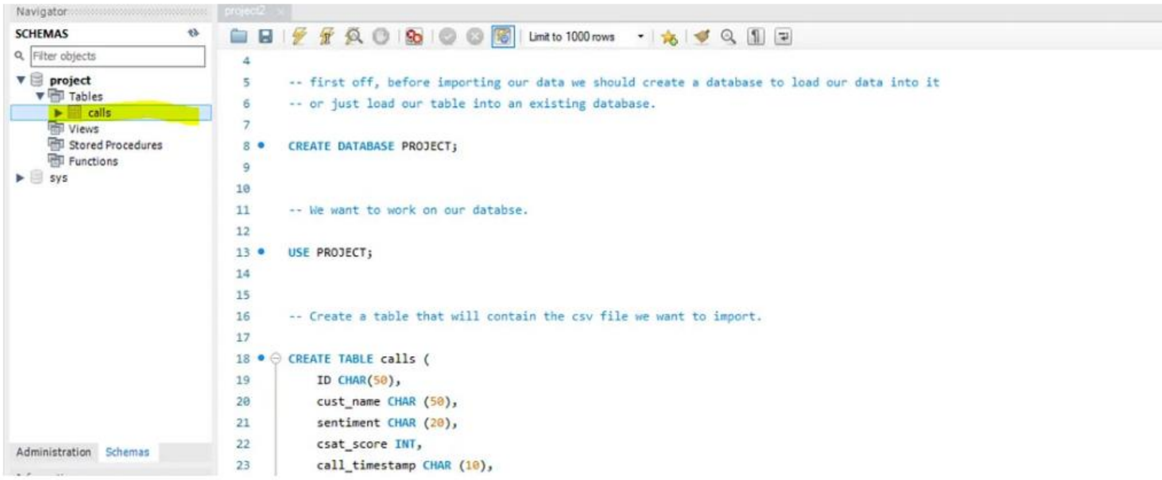
```sql
-- Create a table that will contain the csv file we want to import.

CREATE TABLE calls (
    ID CHAR(50),
    cust_name CHAR (50),
    sentiment CHAR (20),
    csat_score INT,
    call_timestamp CHAR (10),
    reason CHAR (20),
    city CHAR (20),
    state CHAR (20),
    channel CHAR (20),
    response_time CHAR (20),
    call_duration_minutes INT,
    call_center CHAR (20)
);
```

Creating a table.

Here we are creating a table called '*calls*', and we are designing it in a way to fit our data by matching the columns and data types.

PLEASE NOTE: Specifying the size of the variables without checking the CSV file before proved to be stupid. You might lose a lot of important information if the size you specified do not match that of the size in the csv file. Make sure to double check the dataset and SET the appropriate size for each column

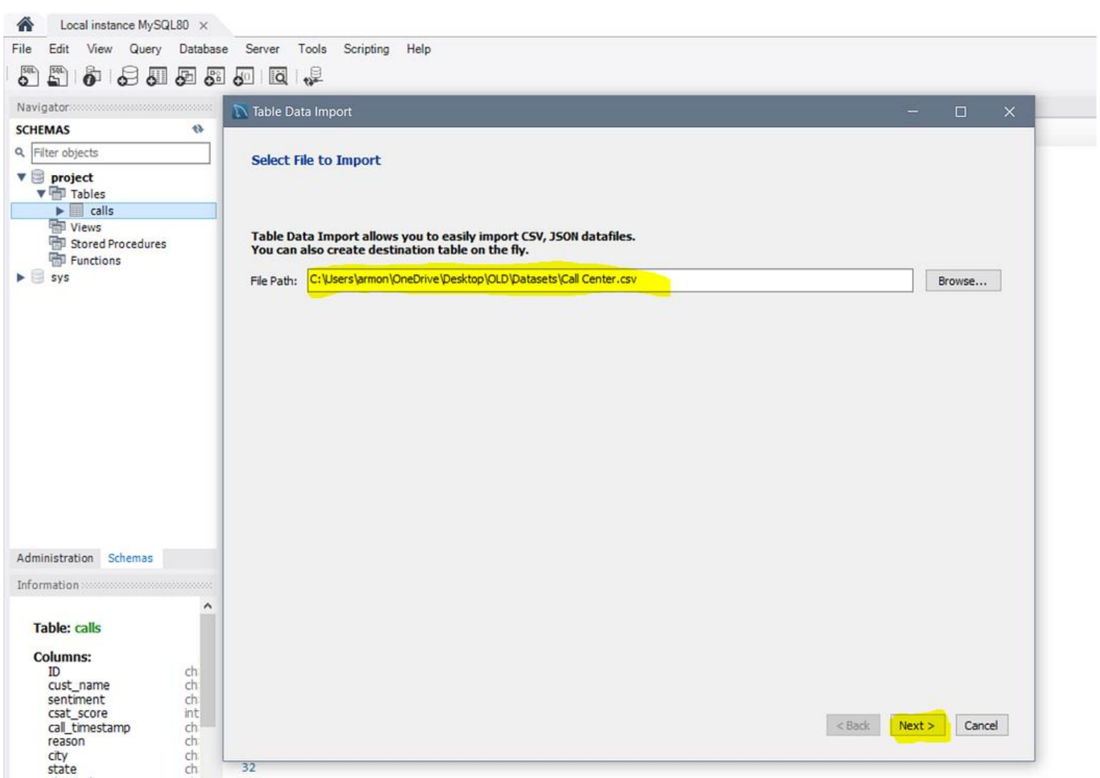
8. Create the table



If you cant find the database or the table, make sure to refresh by pressing the little arrows button next to schemas.

9. IMPORT the DATA

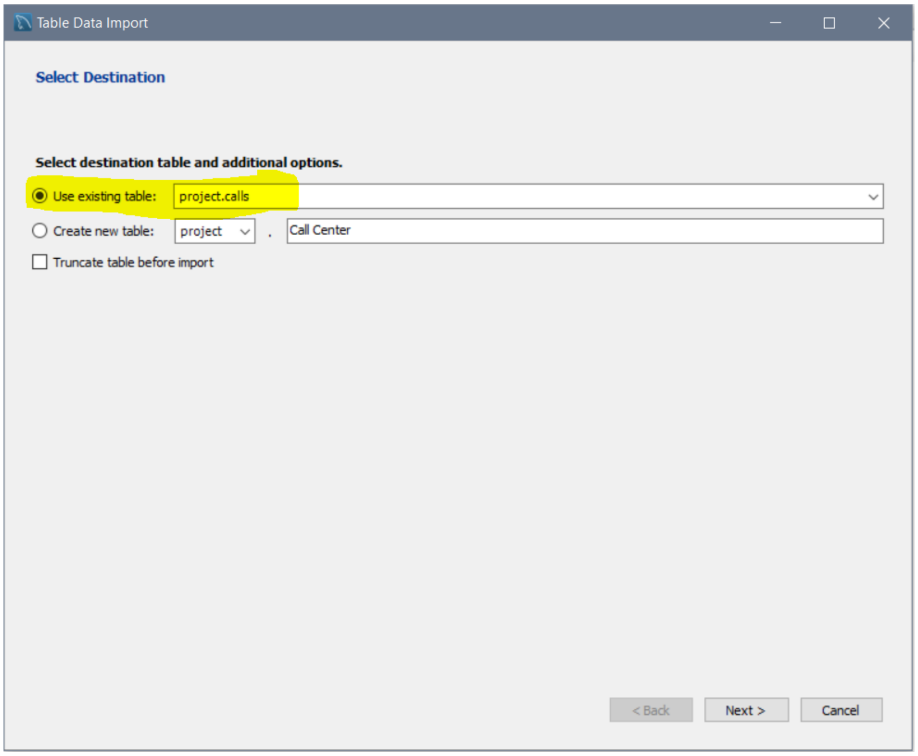
When you right click, there is an option called ‘Table Data Import Wizard’. That is the wizard that will help us ! click on it, it will open a window for you to browse for the csv file you want to import, search for it and then we you finish press next:



specifying the path to our dataset.

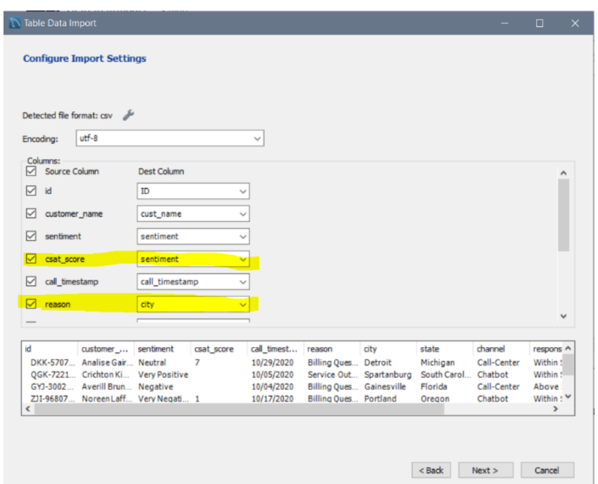
After you press next, it will ask you for the destination where you want the data to go:

Press enter or click to view image in full size



We want the data to go to the table we created so choose the ‘Use existing table’ and then press next.

Now we need to open our eyes very well here, because here we are finalizing the process by making sure everything is good:



As you can see, it shows us the CSV columns and their destination columns in our table. Make sure you select the right column from the dropdown menu

10. To make sure everything went smoothly, let's take a look at our table:

```sql
SELECT * FROM calls LIMIT 10;
```

The table is way too big to see all of it, so let's see first 10 rows. This is the result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

ID	cust_name	sentiment	csat_score	call_timestamp	reason	city	state	channel	response_time	call_duration_minutes	call_center
DKK-57076809-w-055481-fU	Analise Gairdner	Neutral	7	10/29/2020	Billing Question	Detroit	Michigan	Call-Center	Within SLA	17	Los Angeles/CA
QGK-72219678-w-102139-KY	Crichton Kidsley	Very Positive	0	10/05/2020	Service Outage	Spartanburg	South Carolina	Chatbot	Within SLA	23	Baltimore/MD
GYJ-30025932-A-023015-LD	Averill Brundrett	Negative	0	10/04/2020	Billing Question	Gainesville	Florida	Call-Center	Above SLA	45	Los Angeles/CA
ZJI-96807559-i-620008-m7	Noreen Laffina	Very Negative	1	10/17/2020	Billing Question	Portland	Oregon	Chatbot	Within SLA	12	Los Angeles/CA
DDU-69451719-O-176482-Fm	Toma Van der Beken	Very Positive	0	10/17/2020	Payments	Fort Wayne	Indiana	Call-Center	Within SLA	23	Los Angeles/CA
JVI-79728660-U-224285-4a	Kaylyn Emlen	Neutral	5	10/28/2020	Billing Question	Salt Lake City	Utah	Call-Center	Within SLA	25	Baltimore/MD
AZI-95054097-e-185542-PT	Phillipe Bowring	Neutral	8	10/16/2020	Billing Question	Tyler	Texas	Chatbot	Within SLA	31	Baltimore/MD
TWX-27007918-I-608789-Xw	Krysta de Tocqueville	Positive	0	10/21/2020	Billing Question	New York City	New York	Chatbot	Below SLA	37	Los Angeles/CA
XNG-44599118-P-344473-ZU	Oran Lifsey	Very Negative	0	10/03/2020	Billing Question	Dallas	Texas	Email	Below SLA	37	Baltimore/MD
RLC-64108207-Z-285141-VS	Port Inggall	Neutral	0	10/07/2020	Billing Question	Cincinnati	Ohio	Chatbot	Within SLA	12	Baltimore/MD

11. **Step 2: Clean the data.**

Problem 1. Why is the call_timestamp , a date, using a char datatype ? Back in the csv file, the call timestamp was in this format: mm-dd-YYYY. which in human words mean: two digits for month, two digits for day and four digits for year. This is not acceptable in MySQL which it's default format is: yyyy-MM-dd. That is why we made it into a string and then fix it up later in MySQL.

Problem 2. You might've noticed that all the empty values which were in the original/csv dataset that were in the csat_score, got converted to zero's Instead of null values, which will mess up our aggregations because the minimum score is 1 and not 0.

Let's fix them:

1. It's pretty straightforward to be honest, we just call the function str_to_date() and give it our column and the way the date is formatted in it (pay attention here, not the format *we* want, but the format that the string is already in it):

```sql
-- The call_timestamp is a string, so we need to convert it to a date.

SET SQL_SAFE_UPDATES = 0;

UPDATE calls SET call_timestamp = str_to_date(call_timestamp, "%m/%d/%Y");

SET SQL_SAFE_UPDATES = 1;

SELECT * FROM calls LIMIT 10;
```

We need to set the SQL_SAFE_UPDATES off before we do change the column. reason is because we dont specify a where clause that uses a KEY column. That is why we set it off before the query, and then set it back on after. and the results?

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

ID	cust_name	sentiment	csat_score	call_timestamp	reason	city	state	channel	response_time	call_duration_minutes	call_center
DKK-57076809-w-055481-fU	Analise Gairdner	Neutral	7	2020-10-29	Billing Question	Detroit	Michigan	Call-Center	Within SLA	17	Los Angeles/CA
QGK-72219678-w-102139-KY	Crichton Kidsley	Very Positive	0	2020-10-05	Service Outage	Spartanburg	South Carolina	Chatbot	Within SLA	23	Baltimore/MD
GYJ-30025932-A-023015-LD	Averill Brundrett	Negative	0	2020-10-04	Billing Question	Gainesville	Florida	Call-Center	Above SLA	45	Los Angeles/CA
ZJI-96807559-i-620008-m7	Noreen Laffina	Very Negative	1	2020-10-17	Billing Question	Portland	Oregon	Chatbot	Within SLA	12	Los Angeles/CA
DDU-69451719-O-176482-Fm	Toma Van der Beken	Very Positive	0	2020-10-17	Payments	Fort Wayne	Indiana	Call-Center	Within SLA	23	Los Angeles/CA
JVI-79728660-U-224285-4a	Kaylyn Emlen	Neutral	5	2020-10-28	Billing Question	Salt Lake City	Utah	Call-Center	Within SLA	25	Baltimore/MD
AZI-95054097-e-185542-PT	Phillipe Bowring	Neutral	8	2020-10-16	Billing Question	Tyler	Texas	Chatbot	Within SLA	31	Baltimore/MD
TWX-27007918-I-608789-Xw	Krysta de Tocqueville	Positive	0	2020-10-21	Billing Question	New York City	New York	Chatbot	Below SLA	37	Los Angeles/CA
XNG-44599118-P-344473-ZU	Oran Lifsey	Very Negative	0	2020-10-03	Billing Question	Dallas	Texas	Email	Below SLA	37	Baltimore/MD
RLC-64108207-Z-285141-VS	Port Inggall	Neutral	0	2020-10-07	Billing Question	Cincinnati	Ohio	Chatbot	Within SLA	12	Baltimore/MD

Next problem. The zero's in the csat_score. There are two options: either we set them to NULL, or we just leave them be and then when we query the table we just add the clause WHERE csat_score != 0. But I will set them to nulls in this way:

```sql
-- The call_timestamp is a string, so we need to convert it to a date.

SET SQL_SAFE_UPDATES = 0;

UPDATE calls SET call_timestamp = str_to_date(call_timestamp, "%m/%d/%Y");

UPDATE calls SET csat_score = NULL WHERE csat_score = 0;

SET SQL_SAFE_UPDATES = 1;

SELECT * FROM calls2 LIMIT 10;
```

Now let's see the table:

Press enter or click to view image in full size

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

ID	cust_name	sentiment	csat_score	call_timestamp	reason	city	state	channel	response_time	call_duration_minutes	call_center
DKK-57076809-w-055481-fU	Analise Gairdner	Neutral	7	2020-10-29	Billing Question	Detroit	Michigan	Call-Center	Within SLA	17	Los Angeles/CA
QGK-72219678-w-102139-KY	Crichton Kidsley	Very Positive	NULL	2020-10-05	Service Outage	Spartanburg	South Carolina	Chatbot	Within SLA	23	Baltimore/MD
GYJ-30025932-A-023015-LD	Averill Brundrett	Negative	NULL	2020-10-04	Billing Question	Gainesville	Florida	Call-Center	Above SLA	45	Los Angeles/CA
ZJI-96807559-i-620008-m7	Noreen Laffina	Very Negative	1	2020-10-17	Billing Question	Portland	Oregon	Chatbot	Within SLA	12	Los Angeles/CA
DDU-69451719-O-176482-Fm	Toma Van der Beken	Very Positive	NULL	2020-10-17	Payments	Fort Wayne	Indiana	Call-Center	Within SLA	23	Los Angeles/CA
JVI-79728660-U-224285-4a	Kaylyn Emlen	Neutral	5	2020-10-28	Billing Question	Salt Lake City	Utah	Call-Center	Within SLA	25	Baltimore/MD
AZI-95054097-e-185542-PT	Philipe Bowring	Neutral	8	2020-10-16	Billing Question	Tyler	Texas	Chatbot	Within SLA	31	Baltimore/MD
TWX-27007918-I-608789-Xw	Krysta de Tocqueville	Positive	NULL	2020-10-21	Billing Question	New York City	New York	Chatbot	Below SLA	37	Los Angeles/CA
XNG-44599118-P-344473-ZU	Oran Lifsey	Very Negative	NULL	2020-10-03	Billing Question	Dallas	Texas	Email	Below SLA	37	Baltimore/MD
RLC-64108207-Z-285141-VS	Port Inggall	Neutral	NULL	2020-10-07	Billing Question	Cincinnati	Ohio	Chatbot	Within SLA	12	Baltimore/MD

Sweet. Now we finished cleaning, Let's go to Step 3.

Step 3: EDA.

I wonder what is the shape of our table, i.e, the number of columns and rows. Let's see:

```sql
-- ------------------------------------------------------------------------------------------------
-- -------------------------------------------- Exploring our data ---------------------------------------------
-- ------------------------------------------------------------------------------------------------

-- lets see the shape pf our data, i.e, the number of columns and rows

SELECT COUNT(*) AS rows_num FROM calls;
SELECT COUNT(*) AS cols_num FROM information_schema.columns WHERE table_name = 'calls' ;
```

Running the first line gives us:

Result Grid

rows_num
32918

And the second line gives us:

Result Grid

cols_num
12

So we 32,918 records and 12 columns.

```
-- Checking the distinct values of some columns:
```

- `SELECT DISTINCT sentiment FROM calls;`
- `SELECT DISTINCT reason FROM calls;`
- `SELECT DISTINCT channel FROM calls;`
- `SELECT DISTINCT response_time FROM calls;`
- `SELECT DISTINCT call_center FROM calls;`

We are checking the different values possible for the rows we selected. Let's run a random one of these , since they are all similar:

```sql
SELECT DISTINCT sentiment FROM calls;
SELECT DISTINCT reason FROM calls;
SELECT DISTINCT channel FROM calls;
SELECT DISTINCT response_time FROM calls;
SELECT DISTINCT call_center FROM calls;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

call_center
Los Angeles/CA
Baltimore/MD
Denver/CO
Chicago/IL

So looks like we have only 4 call-centres. Let's continue :

```sql
-- The count and precentage from total of each of the distinct values we got:

SELECT sentiment, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
FROM calls GROUP BY 1 ORDER BY 3 DESC;

SELECT reason, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
FROM calls GROUP BY 1 ORDER BY 3 DESC;

SELECT channel, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
FROM calls GROUP BY 1 ORDER BY 3 DESC;

SELECT response_time, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
FROM calls GROUP BY 1 ORDER BY 3 DESC;

SELECT call_center, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
FROM calls GROUP BY 1 ORDER BY 3 DESC;

SELECT state, COUNT(*) FROM calls GROUP BY 1 ORDER BY 2 DESC;
```

To see the distribution of our calls among different columns. Let's see the reason column:

Here we can see that **Billing Questions amount to a whooping 71% of all calls**, with service outage and payment related calls both are 14.4% of all calls.

Moving on, which day has the most calls?

```
96  SELECT DAYNAME(call_timestamp) as Day_of_call, COUNT(*) num_of_calls FROM calls GROUP BY 1 ORDER BY 2 DESC;
97
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Day_of_call	num_of_calls
Friday	5566
Thursday	5479
Wednesday	4445
Tuesday	4407
Saturday	4397
Monday	4332
Sunday	4292

Friday has the most number of calls while Sunday has the least.

Now let’s move to some aggregations:

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```
99   -- Aggregations :
100
101  SELECT MIN(csat_score) AS min_score, MAX(csat_score) AS max_score, ROUND(AVG(csat_score),1) AS avg_score
102  FROM calls WHERE csat_score != 0; # MySql added 0 to blank rows. But the min is 1.
103
104  SELECT MIN(call_timestamp) AS earliest_date, MAX(call_timestamp) AS most_recent FROM calls;
105
106  SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call_duration FROM calls;
107
108  SELECT call_center, response_time, COUNT(*) AS count
109  FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
110
111  SELECT call_center, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
112
113  SELECT channel, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
114
115  SELECT state, COUNT(*) FROM calls GROUP BY 1 ORDER BY 2 DESC;
116
117  SELECT state, reason, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,2,3 DESC;
118
119  SELECT state, sentiment , COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
120
121  SELECT state, AVG(csat_score) as avg_csat_score FROM calls WHERE csat_score != 0 GROUP BY 1 ORDER BY 2 DESC;
122
123  SELECT sentiment, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
124
```

Again, most of them follow the same logic with minor changes. Let’s run a few of them , starting with the one at line 106, querying the min, max and average call duration in minutes:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

min_call_duration	max_call_duration	avg_call_duration
5	45	25.0175

Then let’s check line 108 and 109:

```
108    SELECT call_center, response_time, COUNT(*) AS count
109    FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
110
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

call_center	response_time	count
Baltimore/MD	Within SLA	6846
Baltimore/MD	Below SLA	2765
Baltimore/MD	Above SLA	1388
Chicago/IL	Within SLA	3359
Chicago/IL	Below SLA	1360
Chicago/IL	Above SLA	696
Denver/CO	Within SLA	1738
Denver/CO	Below SLA	692
Denver/CO	Above SLA	343
Los Angeles/CA	Within SLA	8666
Los Angeles/CA	Below SLA	3327
Los Angeles/CA	Above SLA	1738

Here we are checking how many calls are within, below or above the Service-Level - Agreement time. For example we see that Chicago/IL call center has around **3359 calls** Within SLA , and then Denver/CO has 692 calls below SLA. you get it.

At the end I just added one **Window Function** to query the maximum call duration each day and then sort by it.

```
127    SELECT call_timestamp, MAX(call_duration_minutes) OVER(PARTITION BY call_timestamp) AS max_call_duration FROM calls GROUP BY 1 ORDER BY 2 DESC;
128
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

call_timestamp	max_call_duration
2020-10-04	45
2020-10-22	45
2020-10-26	41
2020-10-13	39
2020-10-03	37
2020-10-14	37
2020-10-21	37
2020-10-30	37
2020-10-09	35
2020-10-12	35
2020-10-25	34
2020-10-16	31
2020-10-02	30
2020-10-19	30
2020-10-20	30
2020-10-18	28
2020-10-08	27

Here we see that for example on Oct 4th the maximum call duration was 45 minutes long while on Oct 8th it was 27 minutes long.

That's it for SQL ! You can make as many queries as you want and change to suit your dataset. One thing though, is I wished we had a database with multiple related tables so we can run some joins and more window functions.

Note: You are to Export your Queries and Screenshot of the outputs to be submitted as proof that you did the work

Submit the sql file of your QUERIES and screenshot of your outputs, Post it also it your portfolio